

MATH I

Fall 2026

Week 3

Practical Session Exercises

Engineering Problems on Limits ¶

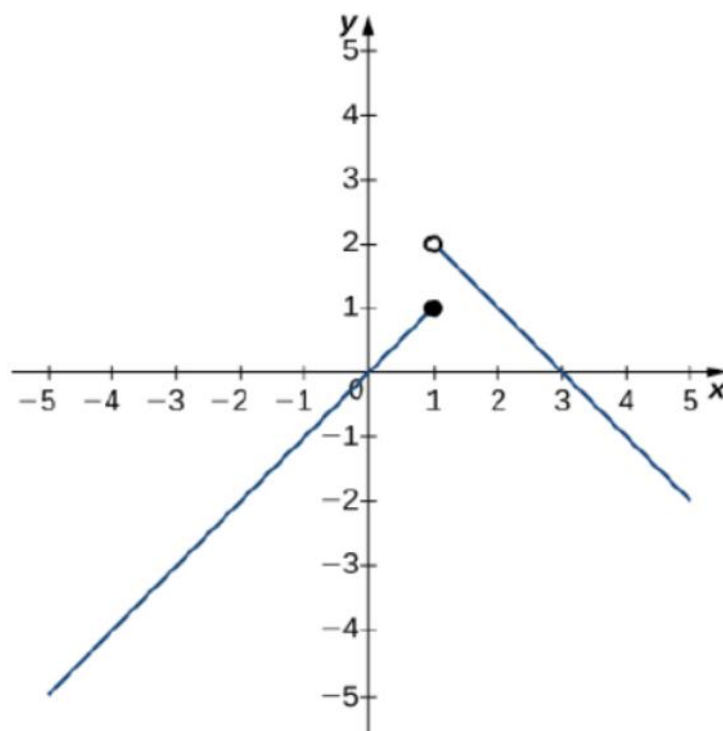
Students should:

- show the mathematical procedure;
- include appropriate physical units (do not forget the units!);
- interpret the result in engineering terms;
- state when a limit does not exist.

Exercise 1

This exercise comprise problems 50-54 from Chapter 1 of textbook.

In the following exercises, use the following graph of the function $y = f(x)$ to find the values, if possible. Estimate when necessary.



51. $\lim_{x \rightarrow 1^+} f(x)$

54. $f(1)$

52. $\lim_{x \rightarrow 1} f(x)$

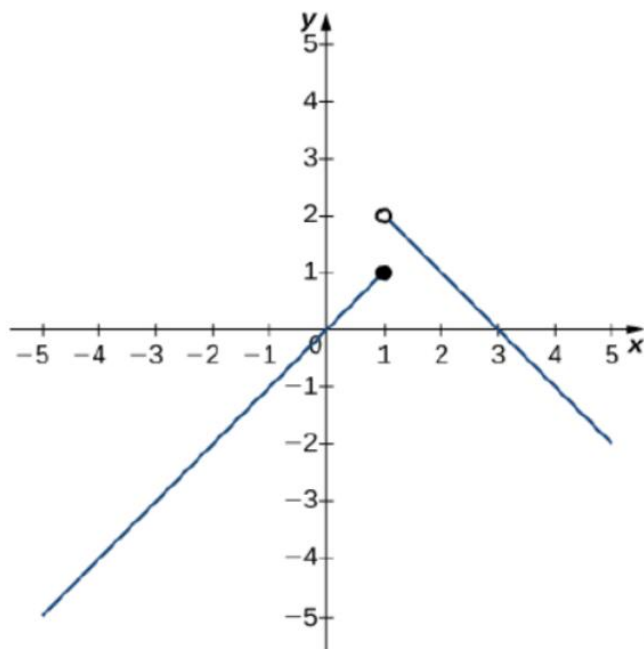
50. $\lim_{x \rightarrow 1^-} f(x)$

53. $\lim_{x \rightarrow 2} f(x)$

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50. $\lim_{x \rightarrow 1^-} f(x)$

53. $\lim_{x \rightarrow 2} f(x)$

54. $f(1)$

Solution

Solution for exercise #50: 1

Solution for exercise #51: 2

Solution for exercise #52: DNE (Does Not Exist)

Solution for exercise #53: 1

Solution for exercise #54: 1

Exercise 2: One-sided limits

Consider

$$f(x) = \begin{cases} 2x + 1, & x < 2, \\ 7 - x, & x \geq 2. \end{cases}$$

Find:

$$\lim_{x \rightarrow 2^-} f(x), \quad \lim_{x \rightarrow 2^+} f(x), \quad \lim_{x \rightarrow 2} f(x), \quad f(2).$$

Explain why the two-sided limit does or does not exist.

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Solution

Values are (respectively) 5, 5, it exists and it is 5 (because the one-sided limits are both equal to 5), 5

Exercise 3 — Calibration near a missing measurement

The output of a temperature sensor is

$$V(T) = \frac{T^2 - 2500}{T - 50}, \quad T \neq 50.$$

1. Explain why $V(50)$ is undefined.
2. Estimate $\lim_{T \rightarrow 50} V(T)$.
3. Explain how the limit could replace the missing calibration value.

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Solution

Direct substitution gives

$$V(50) = \frac{50^2 - 2500}{50 - 50} = \frac{0}{0},$$

so the original formula is undefined at $T = 50^\circ\text{C}$. Factor the numerator:

$$T^2 - 2500 = (T - 50)(T + 50).$$

For $T \neq 50$,

$$V(T) = T + 50.$$

Therefore,

$$\boxed{\lim_{T \rightarrow 50} V(T) = 100 \text{ mV}}.$$

Although the original expression has a hole at $T = 50$, nearby values approach 100 mV. Defining $V(50) = 100 \text{ mV}$ removes the discontinuity and gives a continuous calibration curve.

Exercise 4: A switched control system

The voltage supplied by a controller is

$$V(t) = \begin{cases} 2t + 1, & t < 4, \\ 12 - t, & t \geq 4. \end{cases}$$

Determine:

$$\lim_{t \rightarrow 4^-} V(t), \quad \lim_{t \rightarrow 4^+} V(t), \quad \lim_{t \rightarrow 4} V(t), \quad V(4).$$

Is there an abrupt voltage change at $t = 4$? Explain what this could mean for the controlled device (*think engineering-ly!!*)

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Solution

From the left,

$$\lim_{t \rightarrow 4^-} V(t) = 2(4) + 1 = 9 \text{ V}.$$

From the right,

$$\lim_{t \rightarrow 4^+} V(t) = 12 - 4 = 8 \text{ V}.$$

Since the one-sided limits are unequal,

$$\boxed{\lim_{t \rightarrow 4} V(t) \text{ does not exist.}}$$

Because the second rule includes $t = 4$,

$$\boxed{V(4) = 8 \text{ V}}.$$

The controller commands a sudden drop from values approaching 9 V to 8 V. Such a jump can cause mechanical shock, current transients, or undesirable actuator behaviour.

Exercise 5: Stress–strain behaviour

Introduction: **Stress** describes the internal force acting per unit area within a material when an external load is applied, commonly written as $\sigma = F/A$ and measured in pascals (Pa). **Strain** describes the resulting relative deformation: for axial loading, $\varepsilon = \Delta L/L_0$, and it is dimensionless. The general stress–strain relationship depends on the material and loading conditions and may be written as $\sigma = f(\varepsilon)$. In the linear-elastic region, many materials approximately obey **Hooke’s law**, $\sigma = E\varepsilon$, where E is Young’s modulus; after the elastic limit, the relationship may become nonlinear and involve permanent (plastic) deformation, eventually leading to fracture.

Statement of the problem:

An idealized material has the stress–strain relationship

$$\sigma(\varepsilon) = \begin{cases} 200\,000\varepsilon, & \varepsilon < 0.002, \\ 400 + 10\,000(\varepsilon - 0.002), & \varepsilon \geq 0.002, \end{cases}$$

where stress σ is measured in MPa and strain ε is dimensionless.

1. Find

$$\lim_{\varepsilon \rightarrow 0.002^-} \sigma(\varepsilon).$$

2. Find

$$\lim_{\varepsilon \rightarrow 0.002^+} \sigma(\varepsilon).$$

3. Does the two-sided limit exist?

4. Does the model predict a sudden jump in stress at the transition?

5. Sketch the stress–strain graph near $\varepsilon = 0.002$.

Stress–strain behaviour

$$\sigma(\varepsilon) = \begin{cases} 200\,000\varepsilon, & \varepsilon < 0.002, \\ 400 + 10\,000(\varepsilon - 0.002), & \varepsilon \geq 0.002. \end{cases}$$

Solution

For the elastic branch,

$$\lim_{\varepsilon \rightarrow 0.002^-} \sigma(\varepsilon) = 200\,000(0.002) = 400 \text{ MPa}.$$

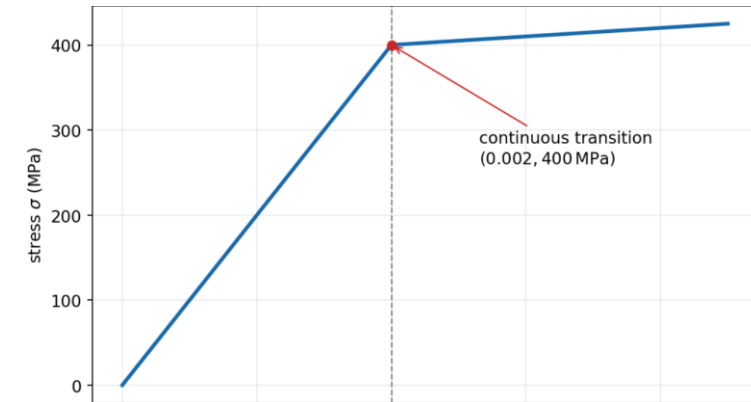
For the second branch,

$$\lim_{\varepsilon \rightarrow 0.002^+} \sigma(\varepsilon) = 400 + 10\,000(0) = 400 \text{ MPa}.$$

The two one-sided limits agree, so

$$\lim_{\varepsilon \rightarrow 0.002} \sigma(\varepsilon) = 400 \text{ MPa}.$$

The model has no stress jump and is continuous at the transition. Its slope changes from 200 000 MPa to 10 000 MPa per unit strain, representing a sharp decrease in tangent stiffness.



Exercise 6: Long-term RC-circuit response

Introduction: A **capacitor** is an electrical component that stores electric charge and energy in the electric field between two conducting plates separated by an insulating material. Its capacitance C relates the stored charge Q to the voltage V by $Q = CV$. When a capacitor is connected to a DC source V_s through a resistor R , it charges gradually rather than instantaneously: the voltage across the capacitor is $V_C(t) = V_s (1 - e^{-t/(RC)})$, assuming it is initially uncharged. Thus, $V_C(0) = 0$, and the voltage approaches V_s asymptotically as $t \rightarrow \infty$; the product $\tau = RC$, called the *time constant*, determines how quickly the capacitor charges.

Statement of the problem:

The voltage across a charging capacitor is

$$V(t) = V_s \left(1 - e^{-t/(RC)}\right),$$

where V_s is the supply voltage.

Let

$$V_s = 12 \text{ V}, \quad R = 10\,000 \, \Omega, \quad C = 100 \, \mu\text{F}.$$

1. Evaluate

$$\lim_{t \rightarrow 0^+} V(t).$$

2. Evaluate

$$\lim_{t \rightarrow \infty} V(t).$$

3. Interpret both limits physically.

Exercise 6 — Long-term RC-circuit response

$$V(t) = V_s \left(1 - e^{-t/(RC)}\right),$$

with $V_s = 12 \text{ V}$, $R = 10\,000 \, \Omega$, and $C = 100 \, \mu\text{F}$.

Solution

Convert the capacitance:

$$C = 100 \times 10^{-6} \text{ F} = 10^{-4} \text{ F}.$$

The time constant is

$$\tau = RC = (10^4)(10^{-4}) = 1 \text{ s}.$$

At switching time,

$$\lim_{t \rightarrow 0^+} V(t) = 12(1 - e^0) = 0 \text{ V}.$$

For long times, $e^{-t/(RC)} \rightarrow 0$, so

$$\lim_{t \rightarrow \infty} V(t) = 12 \text{ V}.$$

The uncharged capacitor initially has zero voltage. It then charges asymptotically toward the supply voltage; in the ideal model it never exceeds **12 V**.

Exercise 7: Terminal velocity

The velocity of a falling object is modelled by

$$v(t) = v_T (1 - e^{-kt}),$$

where $v_T > 0$ is the terminal velocity and $k > 0$.

1. Find $v(0)$.

2. Evaluate

$$\lim_{t \rightarrow \infty} v(t).$$

3. Explain why this model is more realistic than a model in which velocity increases indefinitely (*think with common sense!*).

Exercise 7 — Terminal velocity

$$v(t) = v_T(1 - e^{-kt}), \quad v_T > 0, \quad k > 0.$$

Solution

At $t = 0$,

$$v(0) = v_T(1 - 1) = 0.$$

Since $e^{-kt} \rightarrow 0$ as $t \rightarrow \infty$,

$$\lim_{t \rightarrow \infty} v(t) = v_T.$$

The model incorporates air resistance: acceleration decreases as speed increases, and the object approaches a finite terminal velocity instead of accelerating without bound.

Exercise 8: Data-transfer rate

The effective data-transfer rate of a communication channel is modelled by

$$R(n) = \frac{100n}{n + 20}$$

megabits per second, where n represents the allocated resource level.

1. Calculate $R(10)$, $R(100)$ and $R(1000)$.

2. Evaluate

$$\lim_{n \rightarrow \infty} R(n).$$

3. Explain why increasing n eventually produces diminishing improvements.

4. Identify the maximum theoretical transfer rate predicted by the model.

Exercise 8 — Data-transfer rate

$$R(n) = \frac{100n}{n + 20} \text{ Mbit/s.}$$

Solution

$$R(10) = \frac{1000}{30} \approx 33.33 \text{ Mbit/s,}$$

$$R(100) = \frac{10\,000}{120} \approx 83.33 \text{ Mbit/s,}$$

$$R(1000) = \frac{100\,000}{1020} \approx 98.04 \text{ Mbit/s.}$$

Dividing numerator and denominator by n gives

$$R(n) = \frac{100}{1 + 20/n}.$$

Therefore,

$$\boxed{\lim_{n \rightarrow \infty} R(n) = 100 \text{ Mbit/s}}.$$

The theoretical maximum is 100 Mbit/s. Additional resources yield diminishing gains because the denominator correction $20/n$ becomes small and the rate approaches its capacity asymptotically.